

Internetworking with TCP/IP

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Outline

- Definitions
- Autonomous Systems
- Basic Principles
- Protocols
- Protocol Layering and Encapsulation
- OSI Model
- TCP/IP Model
- Network Organization
- Network Technologies
- Internet Standards
- Useful Addresses

Definitions

- **To connect computers, one needs**

- Hardware, software and, above all, specifications

- **Internetworking**

- Hardware and software technology with the help of which multiple and diverse underlying hardware technologies are interconnected by providing a way to interconnect heterogeneous networks and a set of communication conventions (i.e., software) that make them interoperate

- **internet**

- Multiple networks connected together
- It hides the details of network hardware and permits computers to communicate irrespective of their physical network connections

- **Internet**

- The worldwide internet
- Global virtual network composed by many physical networks
- Internet is NOT a physical network like, e.g., Ethernet

Definitions (cont)

■ Internet

- Global virtual network composed by many physical networks

■ Intermediate System (IS), also known as Interworking Unit (IWU)

- Machine that performs relaying and (perhaps) route discovery functions
- Route discovery: process of finding the best route between two or more networks (or nodes) in the Internet

■ IS types

- Bridges (Data Link Layer)
- Routers (Network Layer)
- Gateways (Transport Layer)

■ IS functions

- Forwarding: using a routing table to make forwarding decisions
- Routing: using route advertisements to acquire the knowledge to create the routing table that the forwarding protocol uses

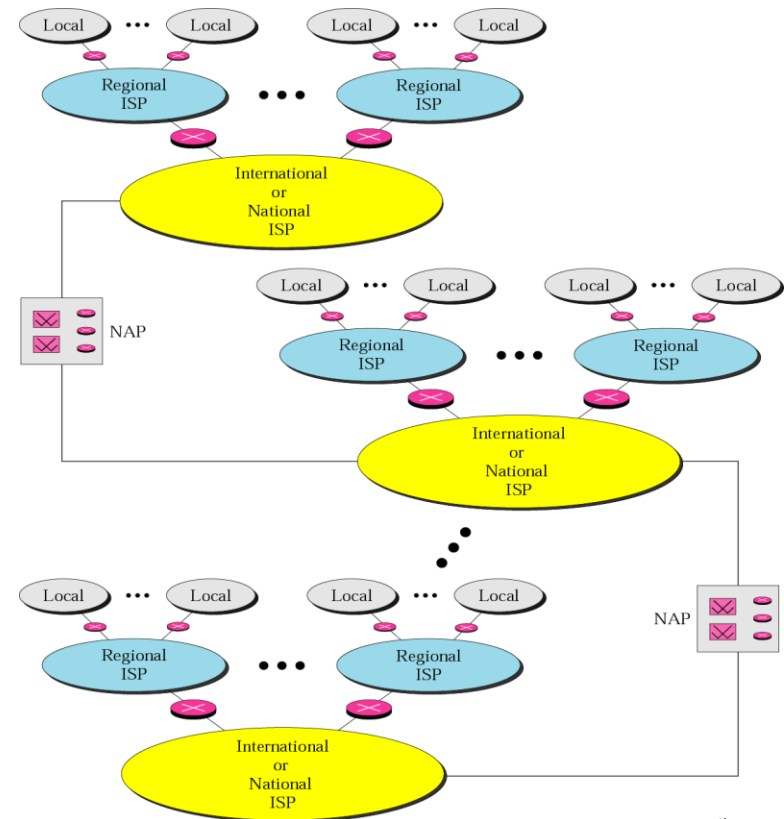


Fig 1.1 – Forouzan 4th edition

Autonomous System

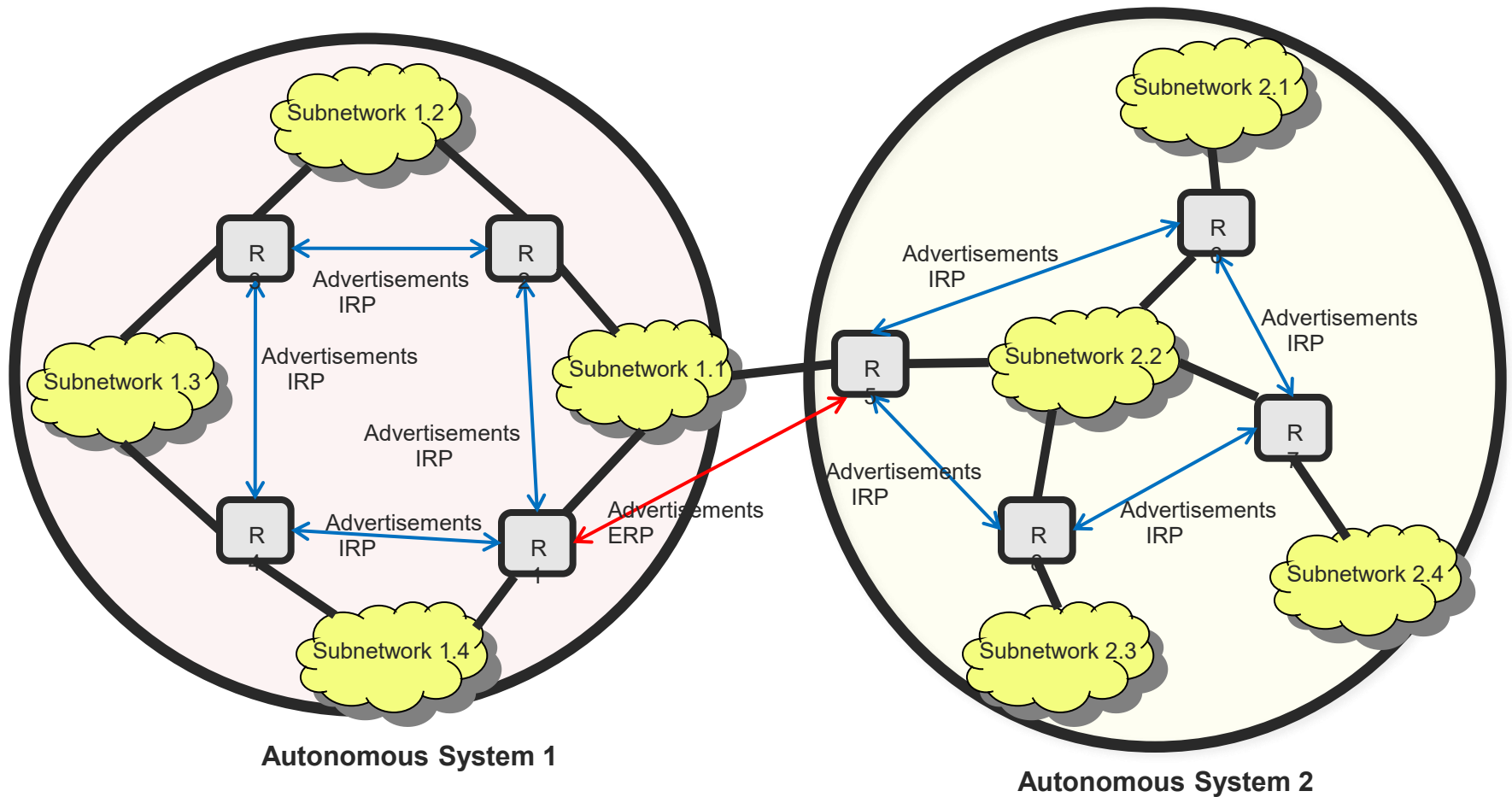
■ Autonomous System (AS)

- Consists of a group of routers connecting several physical networks and exchanging information **via Interior Routing Protocol (IRP)**
- The set of routers and networks are managed by a single organization
- The protocol used to pass routing information among routers placed in different AS is referred to as **Exterior Routing Protocol (ERP)**
- **Three types of AS: Stub AS; Transit AS; Multi-homed AS**

■ Administrative Domain (AD)

- Virtual entity that corresponds to AS
- Administrative entity determined by network administrator (e.g., ISP)
- Functions: to partition the network; security administration; to provide accounting, billing and revenue services; etc.
- Same class of addresses

Autonomous System (cont)



Basic Principles

■ End-to-end principle (David Clark and others)

- Final decisions should always be made by the users
- Trying to implement these decisions by intelligence inside the network is redundant
- Control functions should thus be delegated as much as possible outside the network
- In other words, **the network does the routing and the end-system does the control**

■ IP over everything (Vint Cerf)

- Internet is built by layering a unique “Internetworking protocol” on top of various network technologies
- Consequence: very simple to adopt new technologies in the Internet
- Another advantage: worldwide addressing

■ Connectivity is its own reward

- Connectedness is beneficial to all Internet users
- Exponential growth
- New applications

Protocols

■ Protocol

- Set of rules and conventions used for communication of entities in different systems
- Entity: anything capable of sending or receiving information
- System: object that contains several entities

■ Key elements

- Syntax (data format, coding), e.g., ASCII, EBCDIC, ANS1
- Semantics (meaning, control information)
- Timing (sequencing, synch/speed matching)

■ A protocol specification defines

- Service interface: service offered to users
- Peer interface: how it communicates with other nodes to implement the service interface

■ Communication architecture

- Structured set of protocols that implement the information exchange
- Most of communication architectures are layered

Protocol Layering

■ Protocol layering

- Technique used to partition the tasks of networking into functional layers and assigning protocols to perform each layer's task
- Protocol layering produces simple protocols, each with well-defined tasks
- The protocols can be assembled into a useful entity
- Individual protocols can be removed or replaced as requested for specific applications
- Protocol layering relies on encapsulation and multiplexing/demultiplexing

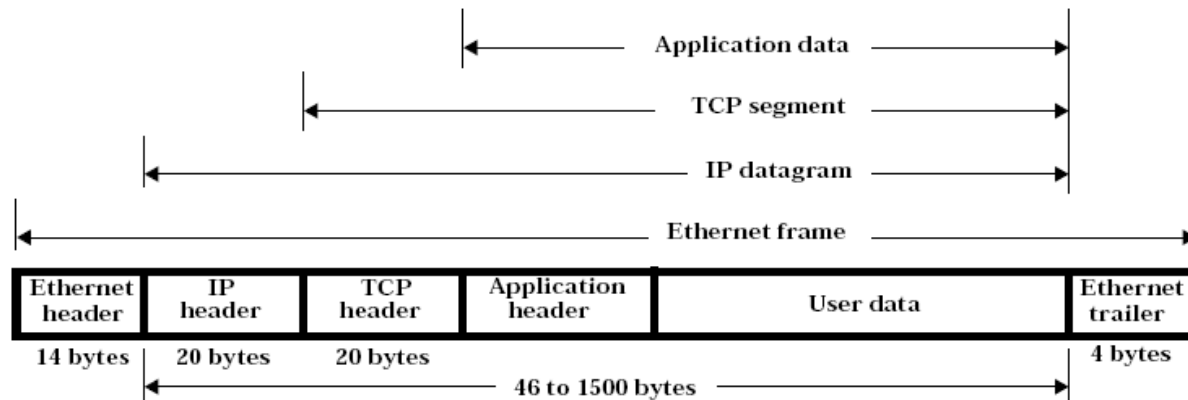
■ Most important layer protocol designs

- OSI model (seven layers)
- DoD (Department of Defense) model (four layers)
- **TCP/IP model (five layers): actual Internet model**
 - Application layer: application specific protocols
 - Transport layer: process to process communication
 - Internet layer: host to host communication
 - Network Interface layer: transmitting IP packets over whatever network technology
 - Hardware: not an official layer

Protocol Encapsulation

■ Protocol encapsulation

- Practice of enclosing data using a specific protocol with messages of another protocol



■ Multiplexing/demultiplexing

- Different techniques of multiplexing and demultiplexing are used throughout the layered hierarchy
- Examples: *destination port number* (application) in TCP/UDP header; *protocol value* (TCP/UDP/ICMP/IGMP) in IP header; *frame type* (IP/ARP/RARP) in Ethernet header

OSI Model

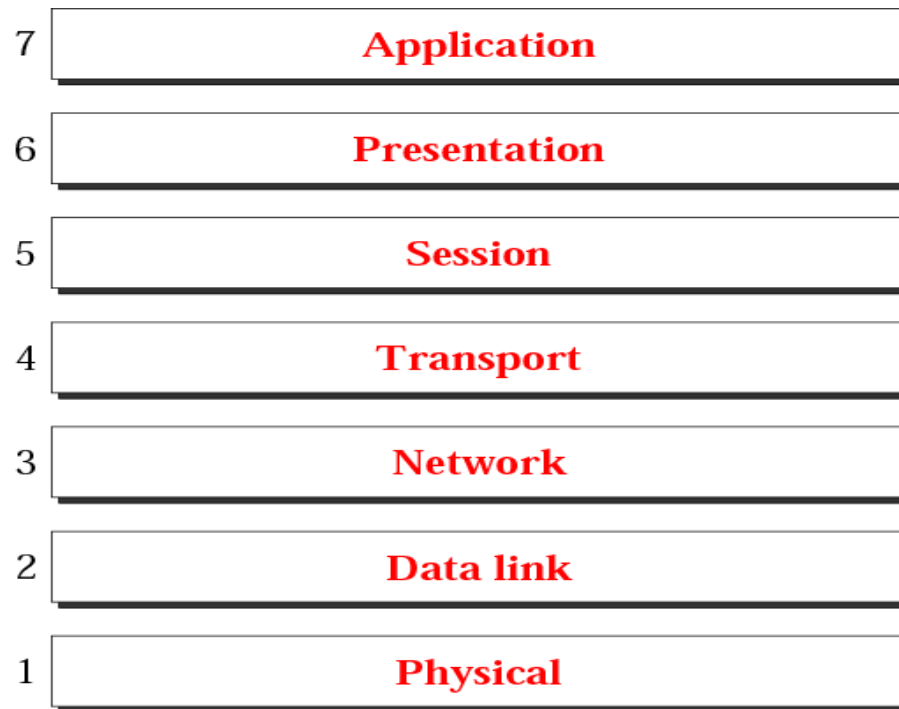


Fig 2.3 – Forouzan 3rd edition

OSI Model (cont)

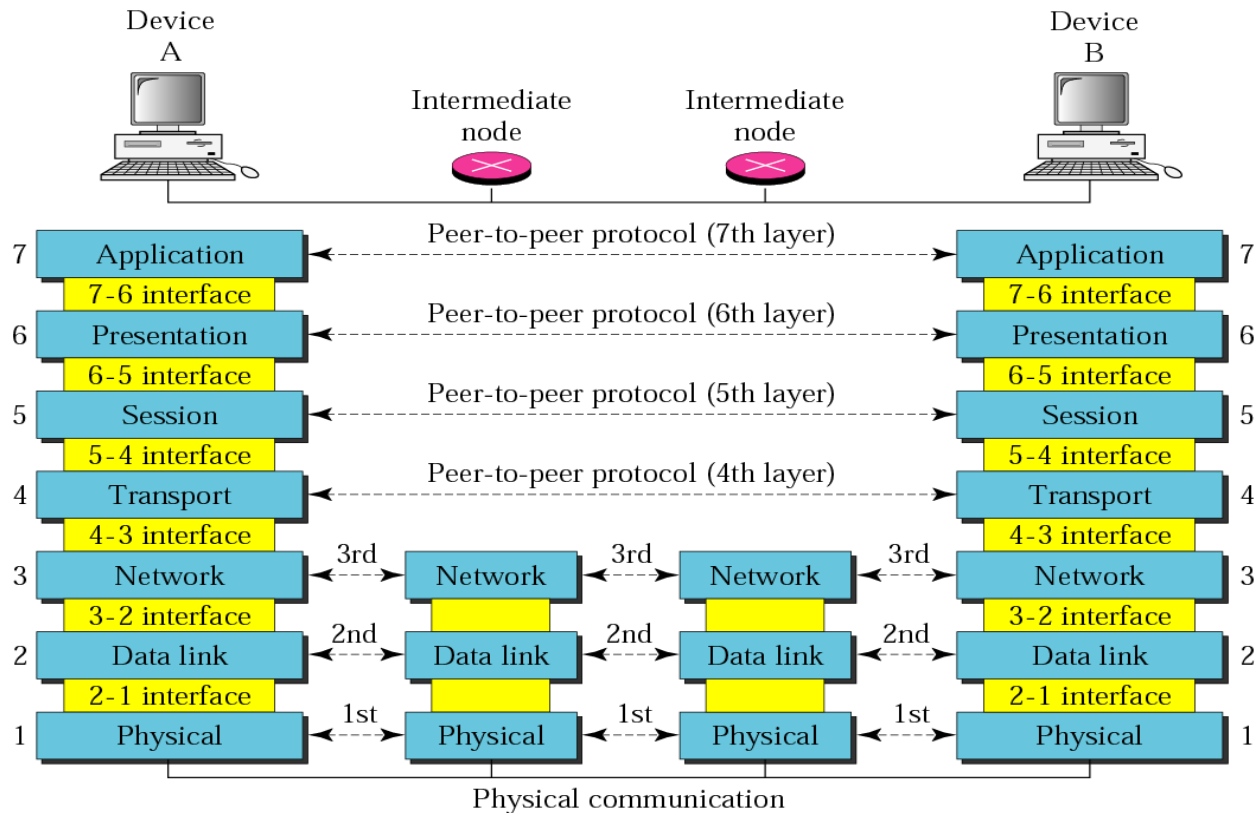


Fig 2.4 – Forouzan 4th edition

OSI Model (cont)

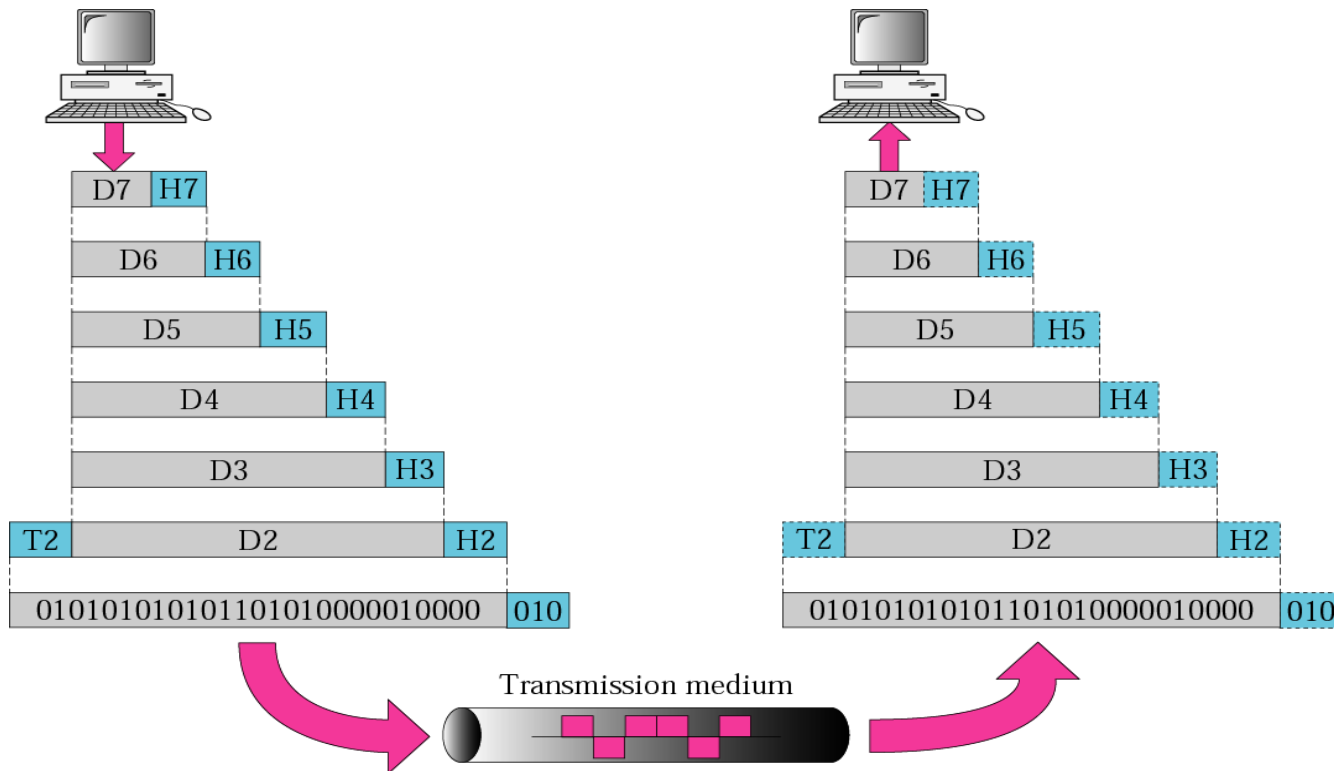


Fig 2.5 – Forouzan 4th edition

OSI Model (cont)

■ Application layer

- **Service:** responsible for the communication management according to the specific service
- **Functions:** everything is application specific
- **Examples:** VT (Virtual Terminal); FTAM (File Transfer, Access and Management); MOTIS (Message Oriented Text Interchange Standard); ASE (Application Specific Service Elements); CASE (Common Application Service Elements); etc
- **Presence at TCP/IP:** yes

■ Presentation layer

- **Service:** provision of common operations on the data structures being exchanged (i.e., freedom from compatibility problems)
- **Functions:** VT support; syntax conversion; cryptography; compression
- **Examples:** ISO presentation protocols; ASN1 (Abstract Syntax Notation 1; universal, for both OSI and TCP/IP)
- **Presence at TCP/IP:** no, it disappears; moved to the Operating System (OS)

■ Session Layer

- **Service:** provision of proper control structure for the specific service, i.e., supports the dialog between cooperating application programs
- **Functions:** session establishment, management and termination (moved to TCP at TCP/IP); synchronization (moved to application at TCP/IP); recovery (moved to OS at TCP/IP)
- **Examples:** RPC (Remote Procedure Call); ISO session protocol
- **Presence at TCP/IP:** no

OSI Model (cont)

■ Transport layer

- **Service:** responsible for the provision of a reliable and transparent data transfer between end users
- **Functions:** connection establishment, management and termination; error control; flow control; message delivery; multiplexing
- **Examples:** ISO TP0-TP4
- **Presence at TCP/IP:** yes

■ Network layer

- **Service:** to make the upper layers independent of the topology, data transmission, routing and switching considerations
- **Functions:** routing and switching functions needed to establish, maintain and terminate switched connections and transfer data between hosts, i.e., routing, addressing, switching and congestion control
- **Examples:** X.25; CLNP (corresponds to IP)
- **Presence at TCP/IP:** yes

■ Data link layer

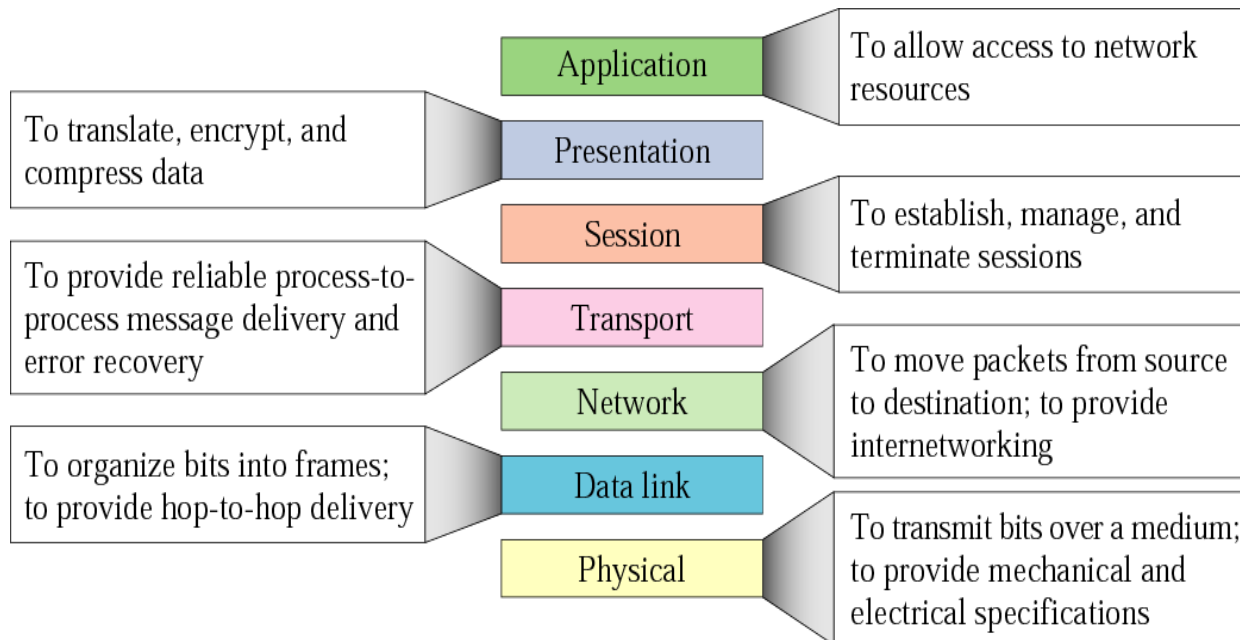
- **Service:** responsible for the reliable transfer of data across the links
- **Functions:** synchronization/framing; error control; flow control
- **Examples:** HDLC; LAP-B; LAP-D
- **Presence at TCP/IP:** yes

OSI Model (cont)

■ Physical layer

- **Service:** responsible for the transmission of bits across the physical medium ensuring reliable delivery of 0's and 1's
- **Functions:** conversion of bits into electrical or optical signals; data transmission
- **Examples:** X.21; RS-232
- **Presence at TCP/IP:** yes

OSI Model (cont)



Summary of OSI layers

Fig 2.6 – Forouzan 4th edition

TCP/IP Model

■ Two models

- **DoD model:** four-layers model developed in the 1970s for the DARPA Internetwork project that grew into the Internet
- **TCP/IP model:** five-layers model that adhere to DoD, although OSI is sometimes preferred for new designs

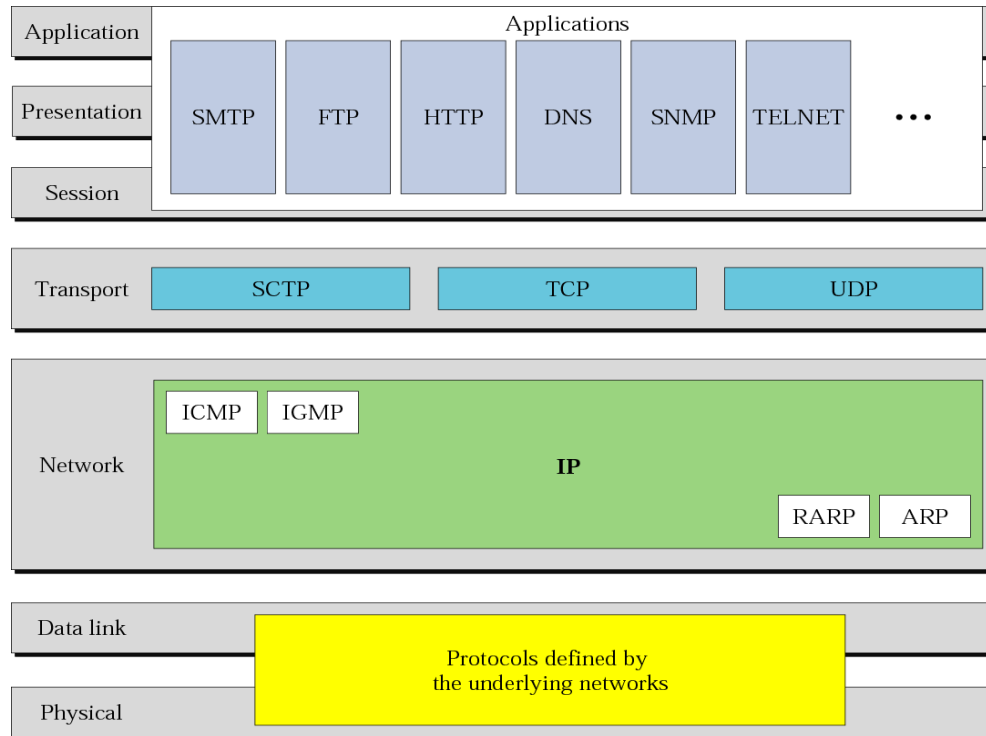
■ DoD (Department of Defense) model

- **Process layer:** protocols implementing user-level functions (corresponds to Application layer in TCP/IP)
- **Host-to-host layer:** handles connection rendezvous, flow control, error control and other generic data flow management (corresponds to TCP in TCP/IP)
- **Internet layer:** responsible for data delivering to destination (routing protocols, corresponds to IP in TCP/IP)
- **Network access layer:** responsible for data delivering over a particular media (corresponds to DLL and PHY in TCP/IP)

■ TCP/IP model

- **Application layer:** HTTP; FTP; Telnet; SMTP; DNS; BGP; DHCP; etc
- **Transport layer:** TCP; UDP
- **Internet layer:** IP
- **Network Interface layer:** CSMA/CD; HDLC; LAP-B; LAP-D;
- **Media/Hardware layer:** Ethernet (twisted pair; broadband coaxial; optical fiber); radio; infrared; etc

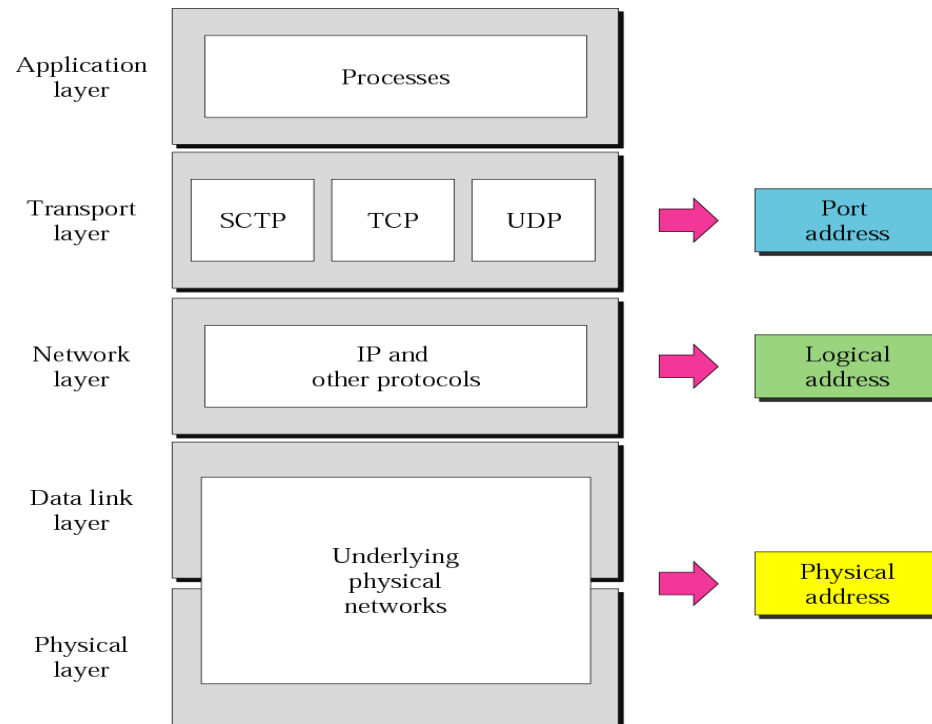
TCP/IP Model (cont)



TCP/IP and OSI model

Fig 2.15 – Forouzan 3rd edition

TCP/IP Model (cont)



Relationship of layers and addresses in TCP/IP

Fig 2.17 – Forouzan 3rd edition

Network Organization

- **Highest level: continental backbone network (continental ingress/egress routers)**
 - Geographically distributed set of very high throughput routers interconnected by very high bit rate lines leased from one or more network providers
- **Same level: core backbone network (continental routers)**
 - The continental backbones are interconnected by means of a core backbone network consisting of a small number of very high throughput (intercontinental) routers and leased lines
- **Second level: regional and/or national networks (national routers)**
 - Geographically distributed set of high throughput routers interconnected by high bit rate leased lines; also, some legacy networks operating a different protocol stack from TCP/IP
- **Next level: access networks (national ingress/egress routers)**
 - Networks attached to one of the regional/national networks/internetworks by means of either an IP access network gateway or the so-called multiprotocol gateway (to access networks with protocol stacks different from TCP/IP), e.g., LAN, diverse ISP networks, enterprise networks
- **Lowest level: hosts**

Network Technologies

■ Types of communication networks

- Telephone Networks (PSTN)
- Local Area Networks (LAN): IEEE 802.3 (CSMA/CD)
- Wireless Local Area Networks (WLAN): IEEE 802.11
- Wide Area Networks (WAN): ADSL; CATV; SONET; FTTH; PPP; X.25; FR; ATM
- Metropolitan Area Networks (MAN)
- Radio Networks (RN): single-hop; multi-hop
- Mobile Communication Networks (MCN): GSM; UMTS
- Satellite Networks (SN)
- Cable Television (CT)
-

■ Ethernet implementations

- 10BASE5; 10BASE2; 10BASE-T; 10BASE-FL;
- Fast Ethernet: 100BASE-TX; 100BASE-FX; 100BASE-T4;
- Gigabit Ethernet: 1000BASE-SX/LX; 1000BASE-T;

■ Different standards for PHY and DLL

■ Important parameter related to network:

Maximum Transmission Unit (MTU)

- MTU depends upon the underlying network
- Expressed in bytes

■ Examples of MTU:

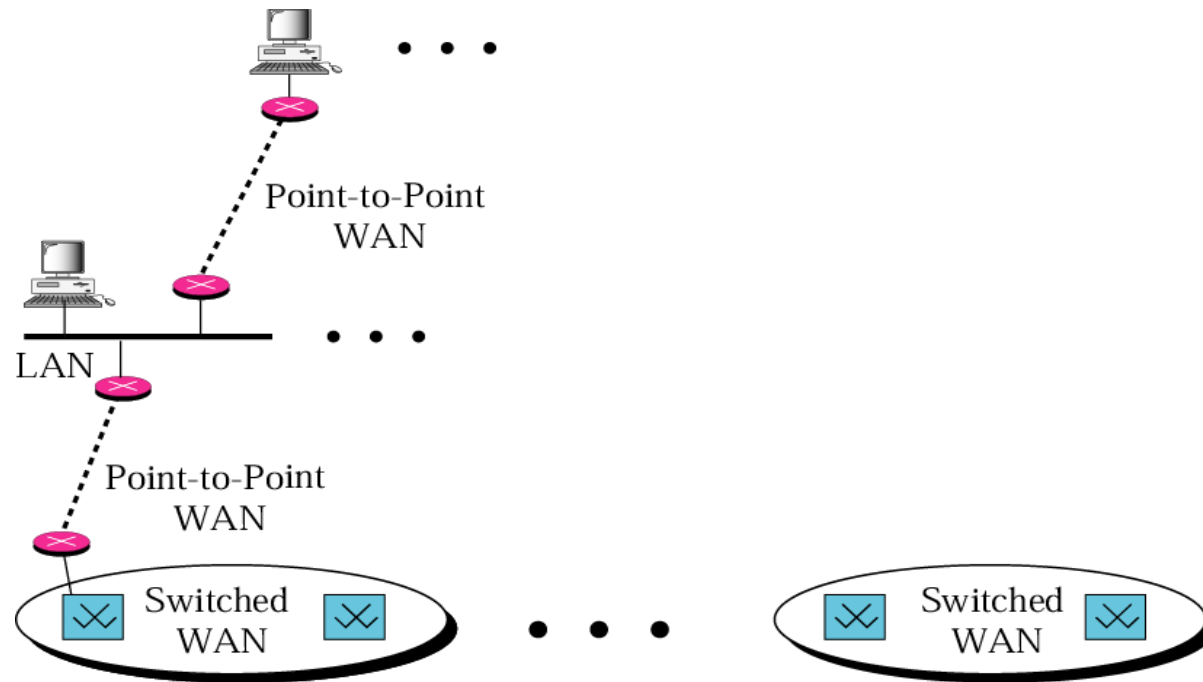
- 1500 (Ethernet IEEE 802.3)
- 8191 (Token Bus IEEE 802.4)
- 4464 (Token Ring IEEE 802.5)
- 4352 (FDDI)
- 576 (X.25)
- 53 (ATM)
-

■ In order to avoid segmentation, one needs to know the so-called **Path MTU**, which is the smallest MTU of the e2e path

■ Besides MTU, there is also a minimum transmission unit, e.g., 64 bytes (Ethernet)

■ IP has also a maximum datagram length (header plus data) of 65,536 bytes

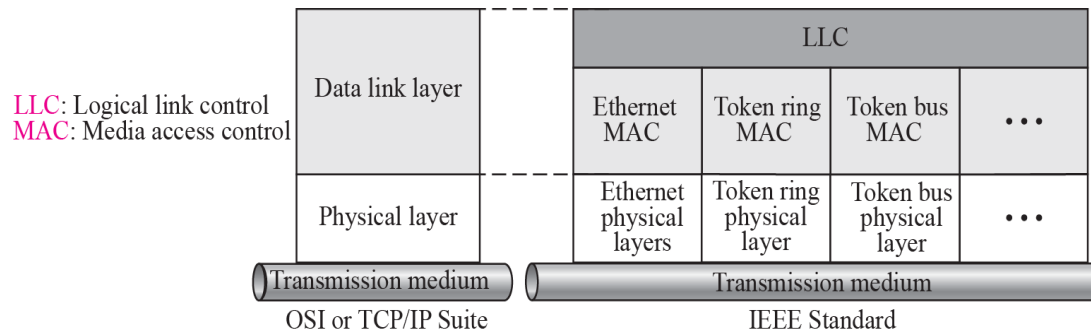
Network Technologies (cont)



Internet model

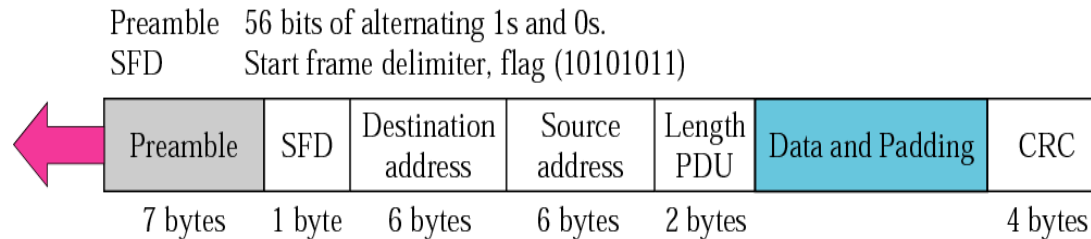
Fig 3.1 – Forouzan 3rd edition

Network Technologies (cont)



Ethernet layers

Fig 3.1 – Forouzan 4th edition



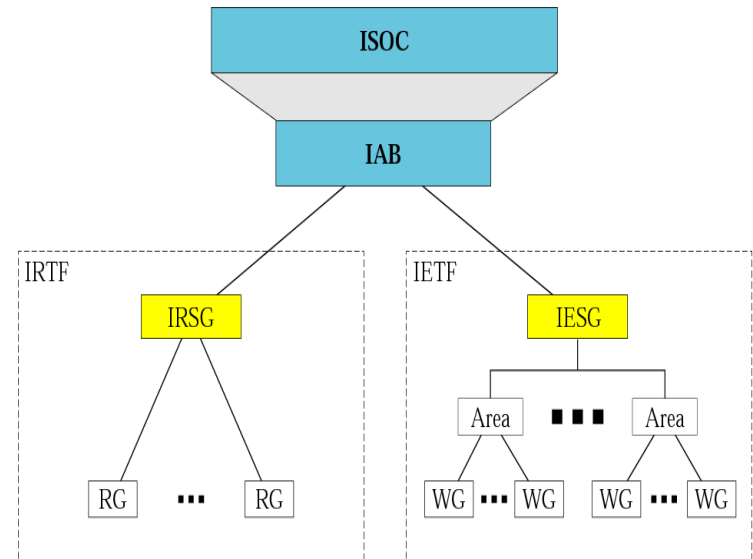
Ethernet frame

Fig 3.2 – Forouzan 4th edition

Internet Standards

■ Groups that coordinate diverse Internet issues

- Internet Society (ISOC)
- Internet Architecture Board (IAB)
- **Internet Engineering Task Force (IETF)**
- Internet Research Task Force (IRTF)
- Internet Engineering Steering Group (IESG)
- Internet Research Steering Group (IRSG)
- **Internet Assigned Numbers Authority (IANA)**
- **Internet Cooperation for Assigned Names and Numbers (ICANN)**
- Network Information Center (NIC)



Internet administration

Fig 1.4 – Forouzan 4th edition

Useful Addresses

- **Internet history**

- <http://www.isoc.org/internet/history/>

- **Internet trends and surveys**

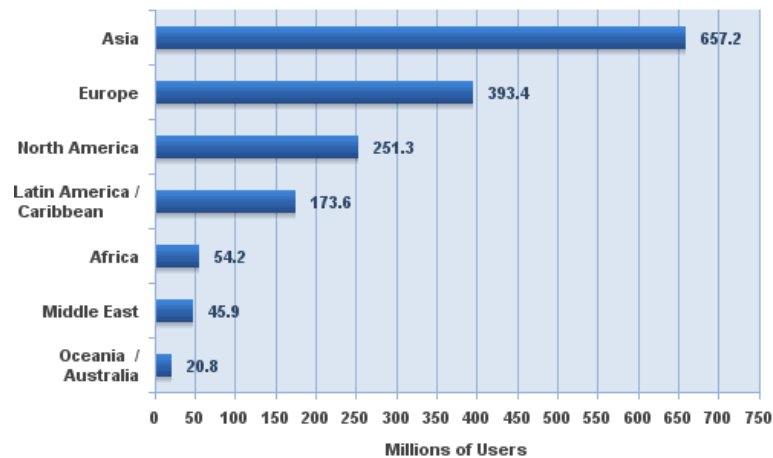
- <http://www.isoc.org/internet/stats/>
- <https://www.isc.org/>
- <http://www.domainstats.com/>
- <http://news.netcraft.com/>
- <http://www.nua.ie/surveys/index.html>
- <http://www.ripe.net/>

- **IETF: Internet specifications**

- <http://www.ietf.org>

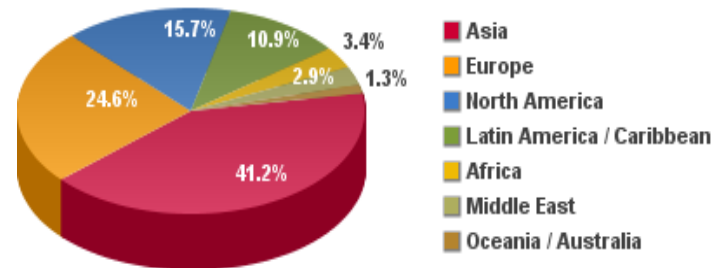
Internet Users

**Internet Users in the World
by Geographic Regions**



Source: Internet World Stats - www.internetworldstats.com/stats.htm
 Estimated Internet users are 1,596,270,108 for March 31, 2009
 Copyright © 2009, Miniwatts Marketing Group

**World Internet Users
by World Regions**



Source: Internet World Stats - www.internetworldstats.com/stats.htm
 1,596,270,108 Internet users for March 31, 2009
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Source: <http://www.internetworldstats.com/stats.htm>
 August 2009

References

- Forouzan B.F, "TCP/IP Protocol Suite", 4th edition, McGraw-Hill, chapters 1, 2, 3
- Other useful addresses

THANK YOU!

